

MATHEMATICS

CONTINUITY AND DIFFERENTIABILITY

Exercise Questions

Continuity Problems

1. Prove $f(x) = 5x - 3$ is continuous at $x = 0, -3, 5$.

2. Examine continuity of $f(x) = 2x^2 - 1$ at $x = 3$.

3. Examine continuity:

(a) $f(x) = x - 5$

(b) $f(x) = \frac{1}{x-5}, x \neq 5$

(c) $f(x) = \frac{x^2 - 25}{x + 5}, x \neq -5$

(d) $f(x) = |x - 5|$

4. Prove $f(x) = x^n$ is continuous.

5.

$$f(x) = \begin{cases} x, & x \leq 1 \\ 5, & x > 1 \end{cases}$$

Check continuity at 0, 1, 2.

6.

$$f(x) = \begin{cases} x + 3, & x \leq 2 \\ x - 3, & x > 2 \end{cases}$$

7.

$$f(x) = \begin{cases} |x| + 3, & x \leq -3 \\ 2x, & -3 < x < 3 \\ 6x - 2, & x \geq 3 \end{cases}$$

8.

$$f(x) = \begin{cases} |x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

9.

$$f(x) = \begin{cases} |x|, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

10.

$$f(x) = \begin{cases} x^2 + 1, & x \geq 1 \\ x + 1, & x < 1 \end{cases}$$

11. Show $g(x) = x - [x]$ is discontinuous at integers.

12. Check continuity of $f(x) = x^2 - \sin x + 5$ at $x = \pi$.

13. Discuss continuity of $\sin x + \cos x$, $\sin x - \cos x$, $\sin x \cos x$.

14. Discuss continuity of cosine, cosecant, secant and cotangent.

15. Determine continuity of

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

16. Show $f(x) = \cos(x^2)$ is continuous.

17. Show $f(x) = |\cos x|$ is continuous.

18. Show $\sin |x|$ is continuous.

19. Find discontinuities of $f(x) = |x| - |x + 1|$.

Differentiation

1. $\sin(x^2 + 5)$

2. $\cos(\sin x)$

3. $\sin(ax + b)$

4. $\sec(\tan x)$

5. $\frac{\sin(ax + b)}{\cos(cx + d)}$

6. $\cos(x^3) \sin^2(x^5)$

7. $(2 \cot x)^2$

8. $\cos(\sqrt{x})$

Non-Differentiability

1. Prove $|x - 1|$ is not differentiable at $x = 1$.
2. Prove greatest integer function $[x]$ is not differentiable at $x = 1$ and $x = 2$.

Implicit Differentiation

Find $\frac{dy}{dx}$:

1. $2x + 3y = \sin x$

2. $2x + 3y = \sin y$

3. $ax + by^2 = \cos y$

4. $xy + y^2 = \tan x + y$

5. $x^2 + xy + y^2 = 100$

6. $x^3 + x^2y + xy^2 + y^3 = 81$

7. $\sin^2 y + \cos(xy) = k$

8. $\sin^2 x + \cos^2 y = 1$